

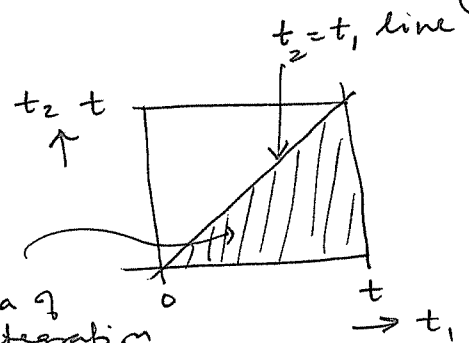
Solution Set 7

①

Consider the integral

$$I = \int_0^t dt_1 \int_0^{t_1} dt_2 f(t_1, t_2)$$

the area of integration



where $f(t_1, t_2)$ is some quantity. The same integral

can be written as

$$I = \int_0^t dt_2 \int_{t_2}^t dt_1 f(t_1, t_2)$$

(If you don't understand this change of integration stop and think about it. It is important to understand that!)

Since t_1 and t_2 are dummy variables we can relabel

$$I = \int_0^t dt_1 \int_{t_1}^t dt_2 f(t_2, t_1)$$

Now

$$I = \frac{1}{2} \left\{ \int_0^t dt_1 \int_0^{t_1} dt_2 f(t_1, t_2) + \int_0^t dt_1 \int_{t_1}^t dt_2 f(t_2, t_1) \right\}$$

Now if we choose

$$f(t_1, t_2) = H(t_1) H(t_2)$$

we get

$$I = \frac{1}{2} \left\{ \int_0^t dt_1 \int_0^{t_1} dt_2 H(t_1) H(t_2) + \int_0^t dt_1 \int_{t_1}^t dt_2 H(t_2) H(t_1) \right\}$$

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Now since t_1 and t_2 always appear in a time ordered way we can replace

$$H(t_1) H(t_2) \rightarrow \hat{T} \{ H(t_1) H(t_2) \} \text{ in 1st integral}$$

$$H(t_2) H(t_1) \rightarrow \hat{T} \{ H(t_1) H(t_2) \} \text{ in 2nd integral}$$

and thus.

$$I = \int_0^t dt_1 \int_0^{t_1} dt_2 H(t_1) H(t_2) = \frac{1}{2} \left\{ \int_0^t dt_1 \int_0^{t_1} dt_2 \hat{T} \{ H(t_1) H(t_2) \} + \int_0^t dt_1 \int_{t_1}^t dt_2 \hat{T} \{ H(t_1) H(t_2) \} \right\}$$

$$= \frac{1}{2} \hat{T} \left\{ \int_0^t dt_1 \hat{H}(t_1) \int_0^t dt_2 \hat{H}(t_2) \right\}$$

$$= \hat{T} \left\{ \frac{1}{2} \left(\int_0^t dt_1 \hat{H}(t_1) \right)^2 \right\}$$

18.2.2 We want to calculate

$$A_{i \rightarrow f} = \left(\frac{-i}{\hbar} \right) \langle 21m | -eE\hat{Z} | 100 \rangle \underbrace{\int_{-\infty}^{\infty} dt e^{i(\omega_{21}t)} e^{-t^2/\tau^2}}_{\sqrt{\pi}\tau^2 e^{-\tau^2\omega_{21}^2/4}}$$

only $l=1$ $m=0$ contributes.

$$P(n=2) = \frac{e^2 E^2}{\hbar^2} \underbrace{|\langle 210 | \hat{Z} | 100 \rangle|^2}_{\frac{2^{15}}{3^{10}} a_0^2} \pi \tau^2 e^{-\tau^2 \omega_{21}^2 / 2}$$

where

$$\omega_{21} = \frac{E_{210} - E_{100}}{\hbar}$$

(see prob 17.2.6 in Set 6)

18.2.3 The velocity of the particle is

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$$v = \frac{\pi \hbar}{mL} \Rightarrow T = \frac{L}{v} = \frac{mL^2}{\pi \hbar}$$

If the box expands symmetrically to twice its size the sudden approximation can be used if $\tau \ll T$.

Thus.

$$\psi_{\text{init}} = \begin{cases} \sqrt{\frac{2}{L}} \cos \frac{\pi x}{L} & -\frac{L}{2} \leq x \leq \frac{L}{2} \\ 0 & \text{outside} \end{cases}$$

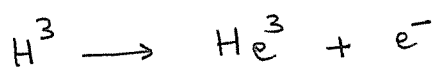
$$\psi_{\text{final}} = \psi_{\text{init}}$$

The wave fn of the new ground state

$$\psi_{\text{init}} = \begin{cases} \sqrt{\frac{1}{L}} \cos \frac{\pi x}{2L} & -L \leq x \leq L \\ 0 & \text{outside} \end{cases}$$

$$\begin{aligned} \therefore \text{Prob} &= \left\{ \int_{-L/2}^{L/2} dx \sqrt{\frac{2}{L}} \left(\cos \frac{\pi x}{2L} \cos \frac{\pi x}{L} \right) \right\}^2 \\ &= \left\{ \frac{1}{\sqrt{2}L} \int_{-L/2}^{L/2} dx \left(\cos \frac{3\pi x}{2L} + \cos \frac{\pi x}{2L} \right) \right\}^2 \\ &= \left\{ \frac{1}{\sqrt{2}L} \left\{ \frac{2L}{3\pi} \left(\sin \frac{3\pi x}{2L} \right)_{-L/2}^{L/2} + \frac{2L}{\pi} \left(\sin \frac{\pi x}{2L} \right)_{-L/2}^{L/2} \right\} \right\}^2 \\ &= \left[\frac{\sqrt{2}}{\pi} \left\{ \frac{2}{3} + 2 \right\} \sin \frac{\pi}{4} \right]^2 = \left(\frac{8}{3\pi} \right)^2 \end{aligned}$$

18.2.4 In the β -decay of H^3



The K.E of e^- is 16 keV.

Mass of the electron 511 keV

The size of H^3 $a_0 = \frac{\hbar^2}{me^2}$ $\frac{v}{c} \approx \sqrt{\frac{16}{511}}$

$\therefore$ Time for the β -particle to leave H^3 is

$$\tau = \frac{\hbar^2}{me^2} \cdot \frac{1}{v} = \frac{\hbar^2}{me^2} \sqrt{\frac{511}{16}} \frac{1}{c}$$

Time period of the electron in H^3

$$T = \frac{\hbar}{me^2} \frac{1}{\alpha c}$$

$$\therefore \frac{\tau}{T} = \sqrt{\frac{511}{16}} \cdot \frac{1}{137} \approx 0.04 \ll 1$$

So we can use the sudden approx.

The Hamiltonian of the Helium atom is

$$H_{He} = \frac{p^2}{2\mu} - \frac{Ze^2}{r} \quad \text{where } Z = 2$$

ground state of He: $\psi_{1,0,0} = \frac{Z^{3/2}}{(\pi a_0^3)^{1/2}} e^{-rZ/a_0}$

ground state of H^3 : $\psi_{1,0,0} = \frac{1}{(\pi a_0^3)^{1/2}} e^{-r/a_0}$

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$$\begin{aligned}
 A_{i \rightarrow f} &= \int d^3r \frac{r^{3/2}}{(\pi a_0^3)} e^{-r(z+1)/a_0} \\
 (\text{amplitude}) &= \frac{4\pi r^{3/2}}{\pi a_0^3} \int_0^\infty dr r^2 e^{-r(z+1)/a_0} \\
 &= \frac{8 r^{3/2}}{(z+1)^3} = \frac{8 \cdot 2\sqrt{2}}{3^3}
 \end{aligned}$$

$\therefore$ Amplitude to be in the ground state = $16\sqrt{2}/27$

Probability to be in some other state $|n\ell m\rangle$ is

$$\psi_{n\ell m} = R_{n\ell}(r, z) Y_{\ell m}(\theta, \phi)$$

So probability to be in $|n=16, \ell=3, m=0\rangle$ state of He^{3+}

is

$${}_{\text{He}} \langle n=16, \ell=3, m=0 | n=1, \ell=0, m=0 \rangle_{\text{H}^3} = 0$$

The $Y_{\ell m}(\theta, \phi)$ integration will give zero!

18.2.6 If $H_1(t) = H_1 \delta(t)$, then we see that-

$$|\psi(t=\frac{\epsilon}{2})\rangle = |\psi(t=-\frac{\epsilon}{2})\rangle + \left(\frac{-i}{\hbar}\right) \int_{-\epsilon/2}^{\epsilon/2} dt H_1 \delta(t) |\psi(t=-\frac{\epsilon}{2})\rangle$$

$$|\psi(t=\frac{\epsilon}{2})\rangle = |i^0\rangle + \left(\frac{-i}{\hbar}\right) H_1 |i^0\rangle$$

$\therefore$ amplitude to go to $|f^0\rangle$ $f \neq i$ is

$$A_{i \rightarrow f} = \left(\frac{-i}{\hbar}\right) \langle f^0 | H_1 | i^0 \rangle$$